

Principles of Communication Systems

Module 2.

Amplitude Modulation Fundamentals:

- AM concepts - Introduction.
- Modulation index. & Percentage of modulation.
- Sidebands and the frequency domain.
- AM power
- Single sideband modulation.

AM Circuits:

- Amplitude Modulator.
- Diode Modulator.
- Transistor Modulator.] Low level AM
- Collector Modulator. → High level AM.
- Amplitude demodulators
- Diode Detector.
- Balanced Modulators.
- Lattice Modulators.

Frequency Division Multiplexing:

- Transmitter - Multiplexer.
- Frequency - Demultiplexer.

Principles of Communication Systems

Module 2:

Amplitude Modulation Fundamentals

AM Concepts:

Introduction:

- * Electronic communication systems can be categorized by the types of information signals transmitted by the system.
- * There are two types of signal:
 - Analog
 - digital.
- * An analog signal is a continuously varying signal, such as a sine wave tone.
 - Eg: voice, video signal.
- * A digital signal has only two distinct levels.
 - ie high & low or ON/OFF.
 - Eg: Data used in computers

Principles of Communication Systems

Modulation:

* Modulation is the process of changing some characteristics (Amplitude, frequency and phase) of the carrier wave in accordance with instantaneous value of the modulating signal.

3 Types of Modulation:

- 1) Amplitude Modulation.
- 2) Frequency Modulation.
- 3) Phase Modulation.

1) Amplitude Modulation: (AM).

* It is defined as the modulation in which the amplitude of the carrier wave is varied in accordance with the instantaneous value of the message signal keeping frequency and phase constant.

Principles of Communication Systems

② Frequency Modulation: (FM)

* It is defined as the Modulation in which the frequency of the carrier wave is varied in accordance with the instantaneous value of the message signal keeping Amplitude and phase constant.

③ Phase Modulation: (PM)

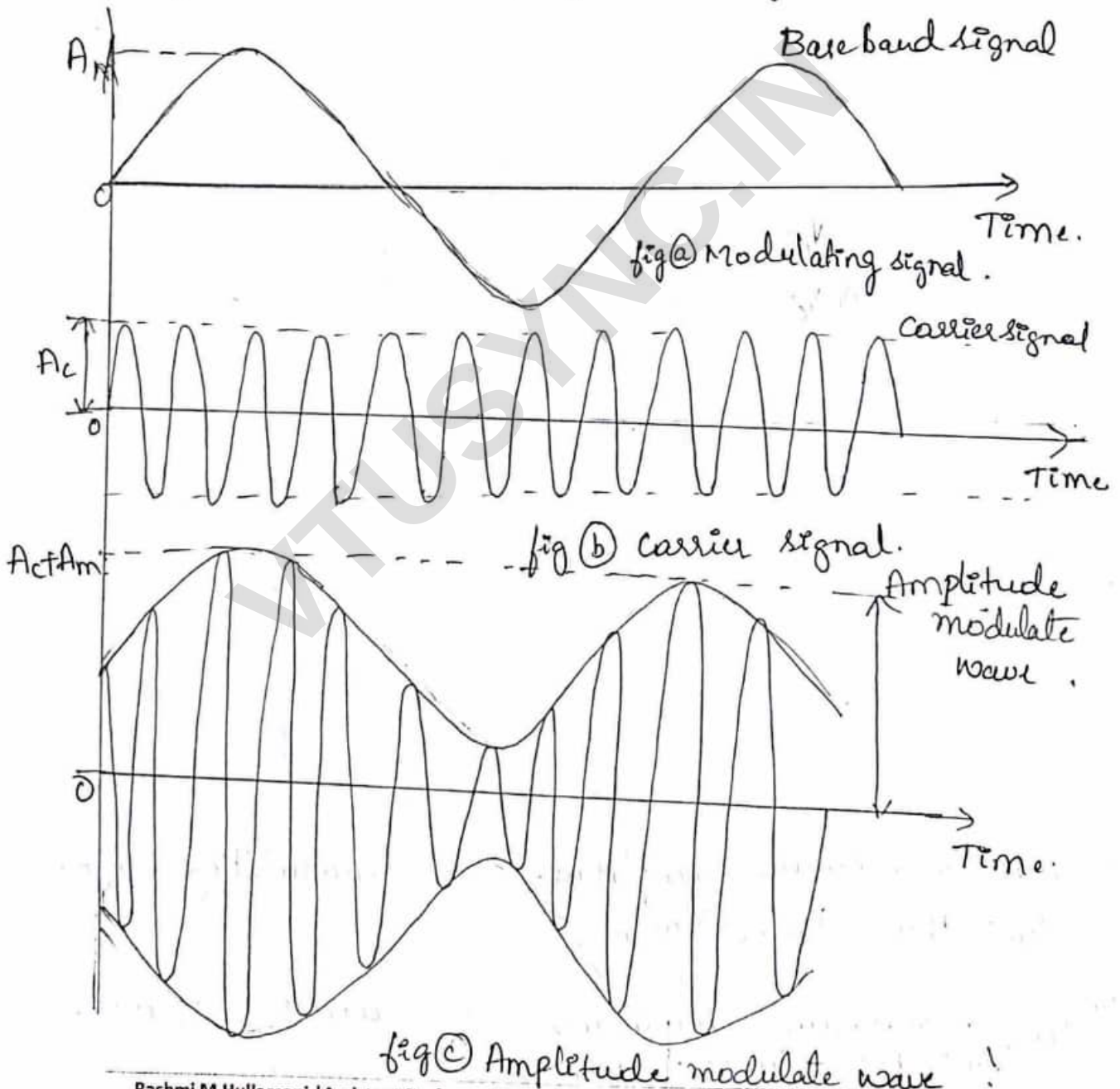
* It is defined as the Modulation in which the phase of the carrier wave is varied in accordance with the instantaneous value of the message signal keeping Amplitude and frequency constant.

Principles of Communication Systems

PRINCIPLES OF COMMUNICATION SYSTEMS -

Amplitude modulation:

- It is defined as the modulation in which the amplitude of the carrier wave is varied accordance with the instantaneous value of the message signal keeping frequency & phase constant.



Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

7

PRINCIPLES OF COMMUNICATION SYSTEMS - I

Time Domain Description:

- The instantaneous values of modulating signal and carrier signal can be represented as Instantaneous value of modulating signal.

$$m(t) = A_m \cos(2\pi f_m t) \quad \text{--- (1)}$$

Instantaneous value of carrier signal.

$$c(t) = A_c \cos(2\pi f_c t) \quad \text{--- (2)}$$

where A_m or V_m = maximum amplitude of modulating signal
 A_c or V_c = maximum amplitude of carrier signal.

The standard form of an amplitude modulated wave is defined by.

$$s(t) = A_c [1 + k_a m(t)] \cos 2\pi f_c t \quad \text{--- (3)}$$

where k_a = constant amplitude sensitivity of the modulator.

The modulated wave so defined is said to be a "standard" AM wave.

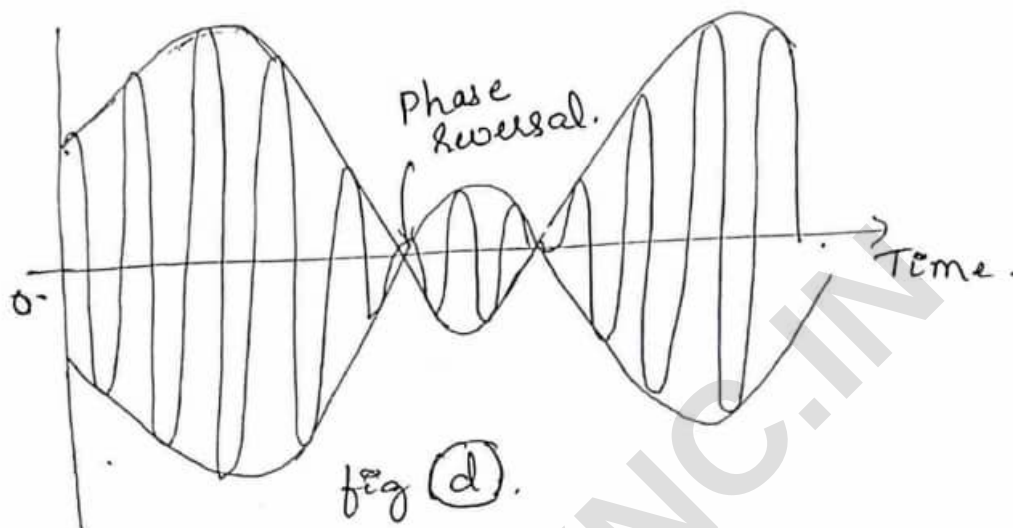
The maximum amplitude of modulating signal
 V_m or $A_m = \frac{V_{\max} - V_{\min}}{2}$

11) The maximum amplitude of carrier signal.
 V_m or $A_c = \frac{V_{\max} + V_{\min}}{2}$

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

PRINCIPLES OF COMMUNICATION SYSTEMS -

- Fig (d) shows amplitude modulated waves for different magnitudes of $k_a m(t)$.



* In fig (c) $k_a m(t)$ is less than 1.

fig (d) $k_a m(t)$ is greater than 1.

When $k_a m(t)$ is greater than 1, the modulated ~~amplitude~~ wave has to suffer from envelope distortion. This wave is called overmodulated amplitude modulate wave.

- For faithful recovery of modulating signal, the non-negative part of envelope must have same shape as the modulating signal $m(t)$. To achieve this, two conditions must satisfy:
- 1) The factor $k_a m(t)$ must be less than 1.
 - 2) The message bandwidth, W , must be small compared to the carrier frequency f_c .

PRINCIPLES OF COMMUNICATION SYSTEMS -

Percentage Modulation or Modulation Factor (μ).

Substituting value of $m(t)$ from (1) in (3).
we get.

$$s(t) = A_c [1 + k_a A_m \cos(2\pi f_m t)] \cos(2\pi f_c t).$$

$$s(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t).$$

where $\mu = k_a A_m$.

where μ is modulation factor or modulation index.

if μ is multiplied by 100, it is
expressed as percentage modulation.

Modulation Index or Modulation factor:

* The ratio of change in Amplitude of modulating signal to the amplitude of carrier wave is known as modulation index (μ).
or modulation factor or modulation Co-efficient
or depth of modulation, or degree of modulation.

$$\boxed{\mu = \frac{A_m}{A_c}} = \mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}.$$

$$\text{or } \mu = \frac{V_m}{V_c}.$$

PRINCIPLES OF COMMUNICATION SYSTEMS -

Frequency Domain Description:

* The modulated carrier has new signals at different frequencies, called side band or side frequencies.

$f_{USB} \rightarrow$ upper side band frequency

$f_{LSB} \rightarrow$ lower side band frequency

$$f_{USB} = f_c + f_m$$

$$f_{LSB} = f_c - f_m$$

* Now consider the ~~instantaneous~~ the amplitude modulated wave.

$$S(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$= A_c \cos(2\pi f_c t) + A_c \underbrace{\mu \cos(2\pi f_m t)}_b \underbrace{\cos(2\pi f_c t)}_a$$

$$\cos a \cos b = \frac{1}{2} [\cos(a-b) + \cos(a+b)]$$

$$= A_c \underbrace{\cos(2\pi f_c t)}_{\text{carrier signal}} + \frac{1}{2} \mu A_c \underbrace{\cos(2\pi f_c - 2\pi f_m)t}_{\text{lower sideband}} + \frac{\mu A_c}{2} \cos(2\pi f_c + 2\pi f_m)t$$

From ① we can say that first term represents unmodulated carrier and two additional terms represent two sideband. ① $\cos(2\pi f_c + 2\pi f_m)t$
upper sideband.

- The frequency of lower sideband $f_{LSB} = f_c - f_m$

- The frequency of upper sideband $f_{USB} = f_c + f_m$ ②

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNCE Shivamogga

PRINCIPLES OF COMMUNICATION SYSTEMS -

Bandwidth of AM wave:

- Bandwidth can be measured by subtracting lowest frequency of the signal from the highest frequency of signal.

- For Amplitude wave (AM) bandwidth is given by.

$$B = f_{USB} - f_{LSB}$$

$$= (f_c + f_m) - (f_c - f_m)$$

$$= f_c + f_m - f_c + f_m$$

$$B = 2f_m = 2W \rightarrow W \text{ is the message bandwidth.}$$

So Bandwidth for AM is ~~define~~ exactly twice the bandwidth of message signal.

PRINCIPLES OF COMMUNICATION SYSTEMS

Power in AM wave:

- AM wave has three components.
 - unmodulated carrier.
 - lower side band.
 - upper side band.

∴ The total power of AM wave is the sum of the carrier power P_c and the power in the two side bands P_{USB} & P_{LSB} .

$$\therefore P_{\text{Total}} = P_c + P_{USB} + P_{LSB}.$$

$$= \frac{V_c^2}{R} + \frac{V_{USB}^2}{R} + \frac{V_{LSB}^2}{R}.$$

$$\begin{aligned} P &= VI \\ I &= V/R \\ P &= \frac{V^2}{R} \end{aligned}$$

① Carrier power:

The carrier power is given by

$$(V = A_c)$$

$$P_c = \frac{A_c^2}{R}.$$

∴ The average carrier power $\frac{(A_c/\sqrt{2})^2}{R}.$

$$\boxed{P_c = \frac{A_c^2}{2R}}$$

PRINCIPLES OF COMMUNICATION SYSTEMS -

⑤ power in sideband:

- Average power for two sidebands can be given.

$$P_{LSB} = P_{USB} = \frac{(V_{SB}/\sqrt{2})^2}{R}$$

$$V_{SB} = \frac{\mu A_c}{2}$$

$$= \left(\frac{\mu A_c / 2}{\sqrt{2}} \right)^2 \times \frac{1}{R}$$

$$= \frac{\mu^2 A_c^2}{\frac{4}{2}} \times \frac{1}{R}$$

$$P_{LSB} = P_{USB} = \frac{\mu^2 A_c^2}{8R}$$

$$\text{wkt} \Rightarrow P_c = \frac{A_c^2}{2R}$$

$$A_c^2 = 2R P_c$$

$$P_{LSB} = P_{USB} = \frac{\mu^2 A_c^2}{8R}$$

$$= \frac{\mu^2 \cancel{2R} P_c}{\cancel{8R} 4}$$

$$P_{LSB} = P_{USB} = \frac{\mu^2 P_c}{4}$$

PRINCIPLES OF COMMUNICATION SYSTEMS -

③ Total power:

$$P_{\text{Total}} = P_c + P_{\text{OSB}} + P_{\text{LSB}}$$

$$= \frac{A_c^2}{2R} + \frac{\cancel{A_c^2} A_c^2}{8R} + \frac{\mu^2 A_c^2}{8R}$$

$$= \frac{A_c^2}{2R} \left[1 + \frac{\mu^2}{4} + \frac{\mu^2}{4} \right] \quad \left(\frac{2\mu^2}{4} \right)$$

$$= \frac{A_c^2}{2R} \left[1 + \frac{\mu^2}{2} \right]$$

$$\boxed{P_{\text{Total}} = P_c \left[1 + \frac{\mu^2}{2} \right]}$$

4) Modulation index in terms of P_{Total} and P_c .

$$P_{\text{Total}} = P_c \left[1 + \frac{\mu^2}{2} \right]$$

$$\frac{P_{\text{Total}}}{P_c} = 1 + \frac{\mu^2}{2}$$

$$\frac{\mu^2}{2} = \frac{P_{\text{Total}}}{P_c} - 1$$

$$\mu^2 = 2 \left[\frac{P_{\text{Total}}}{P_c} - 1 \right]$$

$$\boxed{\mu = \sqrt{2 \left(\frac{P_{\text{Total}}}{P_c} - 1 \right)}}$$

5) Transmission efficiency:

- The transmission efficiency of the AM wave is defined as the ratio of the transmitted power which contains the information to the total transmitted power. (Sum of LSB & USB).

$$\therefore \text{Transmission efficiency } \eta = \frac{P_{\text{LSB}} + P_{\text{USB}}}{P_{\text{Total}}}$$

$$\eta = \frac{\frac{\mu^2 P_c}{4} + \frac{\mu^2 P_c}{4}}{P_c \left[1 + \frac{\mu^2}{2} \right]}$$

$$= \frac{P_c \left[\frac{2\mu^2}{4} \right]}{P_c \left[1 + \frac{\mu^2}{2} \right]}$$

$$\eta = \frac{\frac{\mu^2}{2}}{\frac{2 + \mu^2}{2}} = \frac{\mu^2}{2 + \mu^2}$$

$\therefore$ The percentage transmission efficiency is given by

$$\boxed{\% \eta = \frac{\mu^2}{2 + \mu^2} \times 100\%}$$

PRINCIPLES OF COMMUNICATION SYSTEMS -

Advantages of standard Amplitude modulation.

- 1) AM transmitters are less complex.
- 2) AM receivers are simple, detection is easy.
- 3) AM receivers are not efficient.
- 4) AM waves can travel a longer distance.
- 5) Low bandwidth.

Disadvantages of standard Amplitude modulation

- 1) Amplitude modulation is wasteful of power.
- 2) Amplitude modulation is wasteful of bandwidth.

Applications of AM:

- 1) Radio broadcasting.
- 2) Picture transmission in a TV system.

To overcome these limitations, we can distinguish three modified forms of amplitude modulation:

- 1) Double sideband - suppressed carrier (DSB-SC)
- 2) Vestigial sideband (VSB) modulation.
- 3) Single sideband (SSB) modulation.

PRINCIPLES OF COMMUNICATION SYSTEMS -

Double Sideband Suppressed Carrier Modulation (DSB-SC)

wkt. The amplitude modulated (AM) wave consists of 3 components

- Carrier.
- two sidebands
i.e. upper side & lower side band.

- In the normal AM ~~wave~~ system, both sidebands and carrier are transmitted. The system is commonly known as Double sideband Full carrier system (~~DSB~~ DSB-FC).

- It is important to note that, carrier signal does not convey any information.

∴ two-third of the transmitted power which appears in the carrier is wasted.

- So the real information is conveyed by two sidebands.

- To make the AM wave more efficient we can simply suppress the carrier. Since the carrier does not provide any useful information. When the carrier is removed, the remaining signal contains simply upper ~~sideband~~ & lower sideband such a signal is referred to as a double sideband suppressed

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

(18)
carrier (DSB-SC) signal.

PRINCIPLES OF COMMUNICATION SYSTEMS - 1

Time Domain Description:

Basically, double sideband - suppressed carrier (DSB-SC) modulation consists of the product of the message signal $m(t)$ and carrier wave $c(t)$ as follows:

$$s(t) = c(t)m(t)$$

$$= A_c \cos(2\pi f_c t) m(t)$$

$$= A_c \cos(2\pi f_c t) m(t) \quad \text{--- (1)}$$

This modulated wave undergoes a phase reversal whenever the modulating signal $m(t)$ crosses zero as shown in fig (c).

Frequency Domain Representation:

- The spectrum of modulating signal is shown in fig (a).
- The spectrum of (DSB-SC) modulated wave is shown in fig (b).

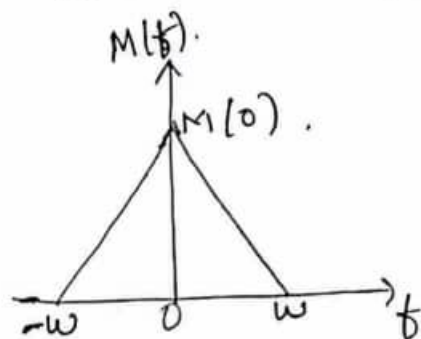


fig (a) Spectrum of modulating signal.

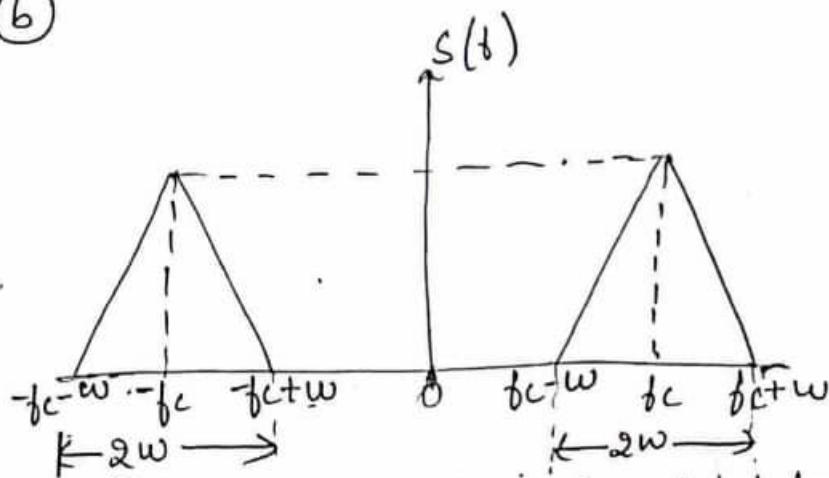


fig (b) Spectrum of DSB-SC modulated wave.

PRINCIPLES OF COMMUNICATION SYSTEMS -

- from fig (b), we can note that the ~~spectrum~~ dotted line indicates that the carrier is suppressed.
- from fig (b) we can note that the spectrum space occupied by a DSB is $2f_m$ or $2W$, which ~~is~~ is same as that ~~of~~ of AM wave.
- The suppression of the carrier from the modulated wave

$s(t) = A_c \cos(2\pi f_c t) m(t)$ can be appreciated by taking its fourier transform.

$$S(f) = \frac{1}{2} A_c [M(f - f_c) + M(f + f_c)]$$

where $S(f)$ - fourier transform of the modulated wave $s(t)$

$M(f)$ - fourier transform of the modulating signal $m(t)$

PRINCIPLES OF COMMUNICATION SYSTEMS -

Single Side Band Suppressed Carrier (SSB-SC) Modulation

- The transmission bandwidth of standard AM as well as DSB-SC modulated wave is $2W$ Hz, i.e. twice the message bandwidth W .

Therefore, both these systems are bandwidth inefficient systems.

- In both these systems, one half of the transmission bandwidth is occupied by the upper sideband (USB) and the other half is occupied by the lower sideband as shown in fig 0

- The information contained in the USB is exactly identical to that carried by the LSB, so by transmitting both the sidebands, we are transmitting the same information twice.

- Hence we can transmit only one sideband (USB or LSB) without any loss of information. So it is possible to suppress the carrier and one sideband completely to conserve the bandwidth.

- When only one side is transmitted, the modulation is called as Single-Sideband Modulation or SSB-SC.

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

(21)

PRINCIPLES OF COMMUNICATION SYSTEMS -

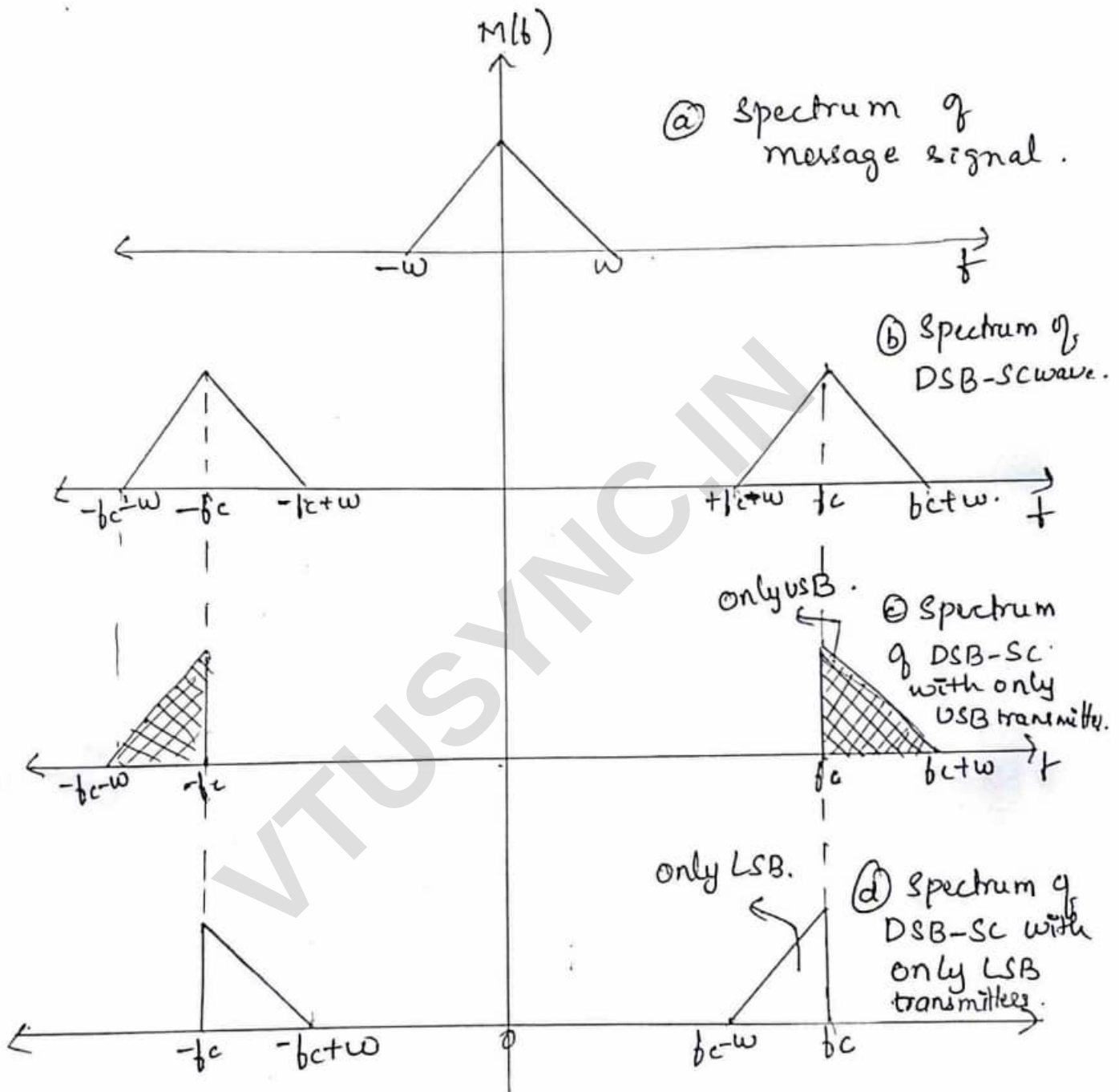


fig ①

Principles of Communication Systems

Problems:

1) Suppose that on an AM signal, the $V_{\max}(P-P)$ value read from the graticule on the oscilloscope screen is 5.9 division and $V_{\min}(P-P)$ is 1.2 divisions.

- a) what is the modulation index.
b) Calculate V_c , V_m and m if the Vertical Scale is 2V division.

Solution:

a) Modulation index?

$$m = \mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

$$\mu = \frac{5.9 - 1.2}{5.9 + 1.2} = 0.662$$

b). $V_c = ?$; $V_m = ?$; $m = ?$

$$V_c = \frac{V_{\max} + V_{\min}}{2} = \frac{5.9 + 1.2}{2}$$

$$= \frac{7.1}{2} = 3.55 \text{ @ } \frac{2V}{\text{div}}$$

$$V_c = 3.55 \times 2V = 7.1V$$

Principles of Communication Systems

$$V_m = \frac{V_{\max} - V_{\min}}{2}$$

$$= \frac{5.9 - 1.2}{2} = \frac{4.7}{2} \text{ @ } \frac{2V}{div}$$

$$V_m = 2.35 \times 2$$

$$V_m = \underline{\underline{4.7V}}$$

$$m = \frac{V_m}{V_c} = \frac{4.7}{7.1} = \underline{\underline{0.662}}$$

② A standard AM broadcast station is allowed to transmit modulating frequencies up to 5 kHz. If the AM station is transmitting on a frequency of 980 kHz, compute the maximum and minimum upper and lower sidebands and the total bandwidth occupied by the AM station.

Solution: given $f_m = 5 \text{ kHz}$,
 $f_c = 980 \text{ kHz}$.

$$f_{USB} = f_c + f_m \\ = 980 + 5 = 985 \text{ kHz}$$

$$f_{LSB} = f_c - f_m \\ = 980 - 5 = 975 \text{ kHz}$$

$$BW = f_{USB} - f_{LSB} = 985 \text{ K} - 975 \text{ K} = 10 \text{ kHz}$$

$$\text{or } BW = 2(f_m) = 2 \times 5 \text{ kHz} = 10 \text{ kHz}$$

(24)

Principles of Communication Systems

- ③ An AM transmitter has a carrier power of 30W. The percentage of modulation is 85 percent. Calculate
- (a) the total power
 - (b) the power in one sideband.

Solution:

Given $P_c = 30 \text{ W}$.
 $\mu = 0.85$

$$(a) P_T = P_c \left[1 + \frac{\mu^2}{2} \right]$$
$$= 30 \left[1 + \frac{(0.85)^2}{2} \right]$$

$$P_T = 40.8 \text{ W}$$

$$(b) P_{LSB} = P_{USB} = \frac{\mu^2 P_c}{4}$$
$$= \frac{(0.85)^2 \times 30}{4} = \underline{\underline{5.4 \text{ W}}}$$

Principles of Communication Systems

(4) An antenna has an impedance of 40Ω . An unmodulated AM signal produces a current of $4.8A$. The modulation is 90 percent. Calculate

- (a) the carrier power.
- (b) the total power.
- (c) the sideband power.

Solution:

$$R = 40\Omega, I = 4.8A, m = 0.90.$$

(a) carrier power:

$$P_c = I^2 R = (4.8)^2 \times 40 = \underline{\underline{921.6W}}$$

$$(b) P_T = P_c \left[1 + \frac{m^2}{2} \right]$$
$$= 921.6 \left[1 + \frac{(0.90)^2}{2} \right]$$

$$P_T = \underline{\underline{1295W}}$$

$$(c) P_{USB} = P_{LSB} = \frac{m^2 P_c}{4} = \frac{(0.90)^2 \times 921.6}{4}$$
$$= \underline{\underline{186.6W}}$$

Principles of Communication Systems

AM Circuits:

* Amplitude modulators are general one of two types.

- 1) Low level.
- 2) High level.

1) Low level:

Low level modulators generate AM with small signals and thus must be amplified considerably if they are to be transmitted.

2) High level:

High level modulators produce AM at high power levels, usually in the final amplifier stage of a transmitter.

Low level Amplitude modulators:

- 1) Diode Modulator.
- 2) Transistor Modulator.

High level Amplitude Modulators:

- 1) collector modulator.

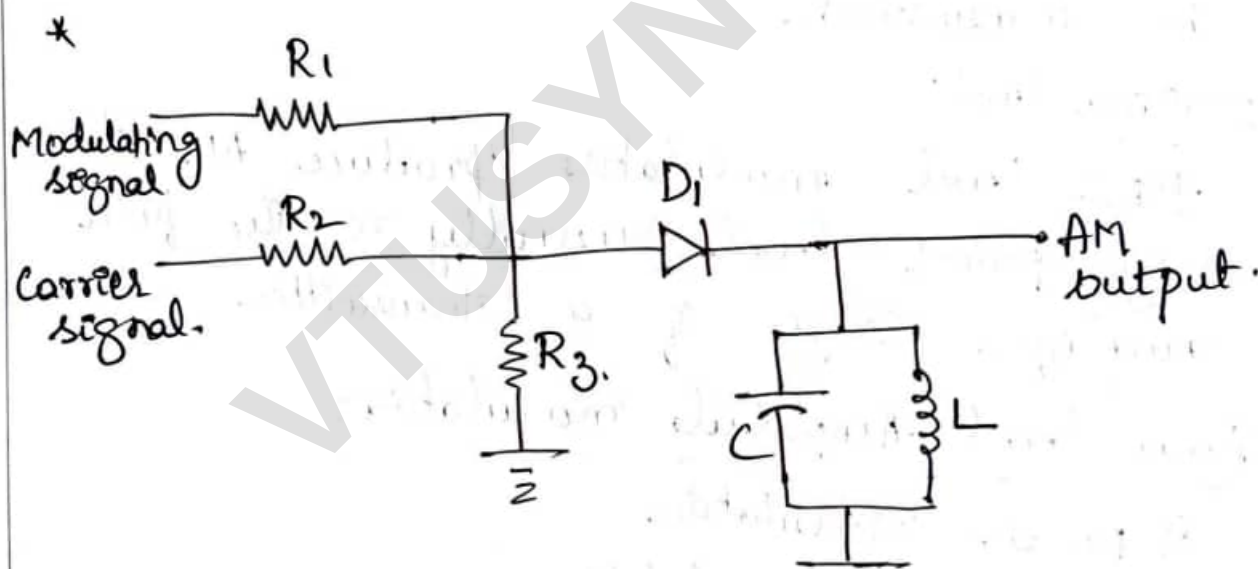
Principles of Communication Systems

Low-level AM:

1) Diode Modulator:

* One of the simplest amplitude Modulator is the diode modulator.

* The diagram for Diode modulator is as shown in figure, which consists of -
- resistive mixing network
- a diode rectifier
- an LC tuned circuit.



* The carrier signal is applied to the one input resistor ~~and~~ and the waveform is shown in fig (a)

* The modulating signal to the other $\rightarrow$ fig (b)

* The mixed signals appear across R_3 .

Principles of Communication Systems

- * This network causes the two signals to be linearly mixed i.e. algebraically added.
- * If both the carrier and the modulating signal are sine waves, the waveform resulting at the junction of the two resistors will be taken that shown in fig (c), where the carrier wave is riding on the modulating signal. This signal is not AM. Modulation is a multiplication process, not an addition process.
- * The composite waveform is applied to a diode rectifier. The diode is connected so that it is forward-biased by the positive-going half cycle of the input wave. During the negative portions of the wave, the diode is cut off and no signal passes.
- * The current through the diode is a series of positive-going pulses whose amplitude varies in proportion to the amplitude of the modulating signal which is shown in fig (d)

Principles of Communication Systems

- * These positive-going pulses are applied to the parallel-tuned circuit made up of L and C , which are resonant at the carrier frequency.
- * Each time the diode conducts a pulse of current flows through the tuned circuits.
- * The coil and capacitor repeatedly exchange energy, causing an oscillation or ringing at the resonant frequency.
- * The oscillation of the tuned circuit creates one negative half-cycle for every positive input pulse. High amplitude positive pulses cause the tuned circuit to produce high-amplitude negative pulses. Low amplitude positive pulses produce corresponding low-amplitude negative pulses.
- * The resulting waveform across the tuned circuit is an AM signal as shown in fig (e)

Principles of Communication Systems

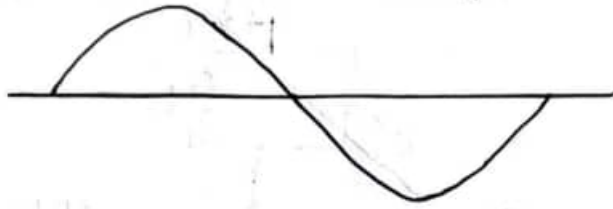


fig (a)
Modulating
signal

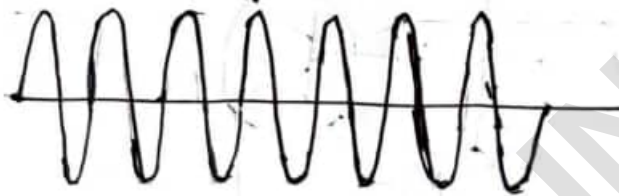


fig (b)
Carrier signal



fig (c)
Linearly mixed
modulating
signal and
Carrier



fig (d)
Positive-going
signal after
diode D_1

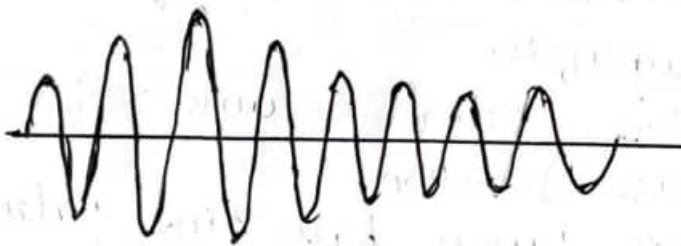
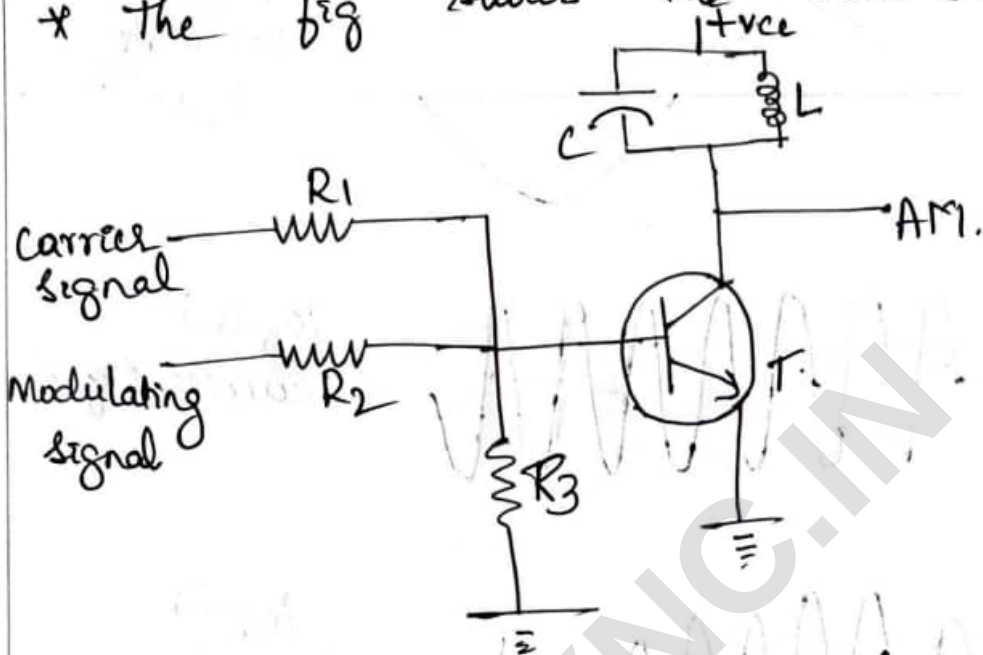


fig (e)
AM output
signal.

Principles of Communication Systems

Transistor modulator: (low level AM)

* The fig shows the transistor modulator



- * It uses transistor instead of the diode, the circuit has gain.
- * The emitter-base junction is a diode and a nonlinear device.
- * The base current controls a larger collector current, and therefore the circuit amplifies.
- * Rectification occurs because of the emitter-base junction.
- * This causes larger half-cycle pulses of current in the tuned circuit.
- * The tuned circuit oscillates to generate

the missing half-cycle. The o/p is a classic AM wave.

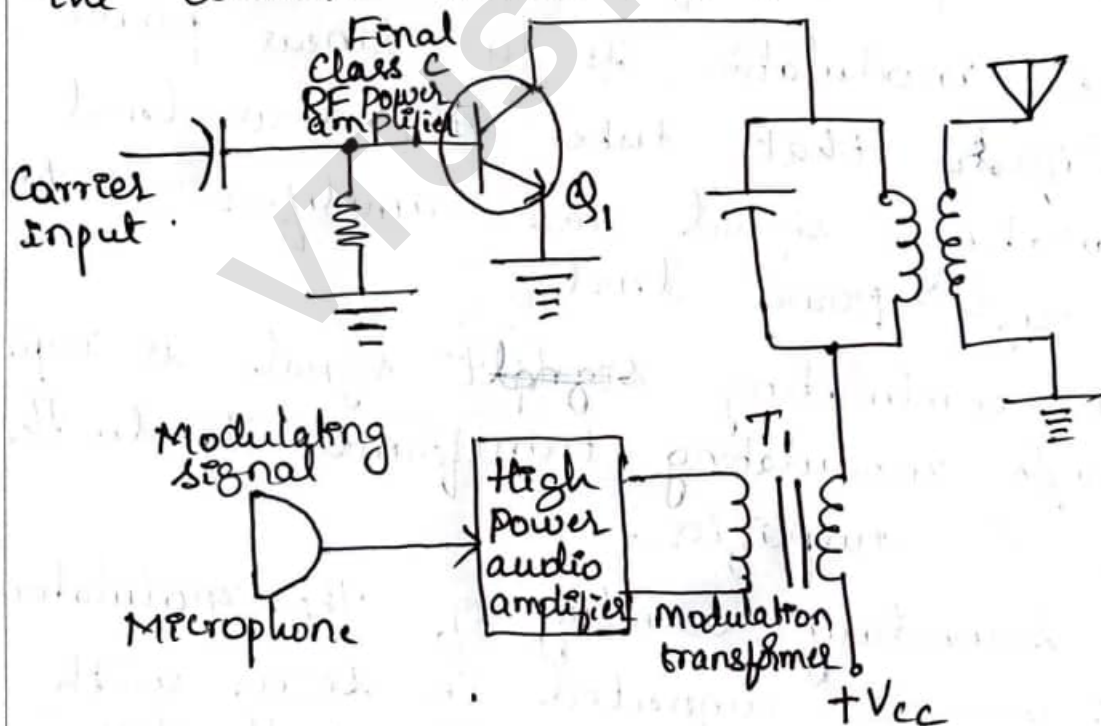
Principles of Communication Systems

High level AM;

- * In high-level AM, the modulator varies the voltage and power on the final RF amplifier stage of the transmitter.
- * The result is high efficiency in the RF amplifier and overall high quality performance.

collector modulator: (High level AM).

- * The high level modulator circuit is the collector modulator as shown in fig.



Principles of Communication Systems

- * The output stage of the transmitter is a high-power class C amplifier.
- * Class C amplifier conducts for only a portion of the positive half-cycle of their input signal.
- * The collector current pulses cause the tuned circuit to oscillate at the desired output frequency.
- * The tuned circuit, reproduces the negative portion of the carrier signal.
- * The modulator is a linear power amplifier that takes the low-level modulating signal and amplifies it to a high-power level.
- * The modulating output signal is coupled through modulation transformer T_1 to the class C amplifier.
- * The secondary winding of the modulation transformer is connected in series with the collector supply voltage V_{CC} of the class C amplifier.

Principles of Communication Systems

- * with a zero-modulation input signal, there is zero-modulation voltage across the secondary of T_1 ; the collector supply voltage is applied directly to the class C amplifier, and the output carrier is a steady sine wave.
- * when the modulating signal occurs, the ac voltage of the modulating signal across the secondary of the modulation transformer is added to and subtracted from the dc collector supply voltage.
- * This varying supply voltage is then applied to the class C amplifier, causing the amplitude of the current pulses through transistor Q_1 to vary.
- * As a result, the amplitude of the carrier sine wave varies in accordance with the modulated signal.
- * when the modulating signal goes positive, it adds to the collector supply voltage, thereby increasing its value and causing

Principles of Communication Systems

- higher current pulses and a higher-amplitude carrier
- * when the modulating signal goes negative, it subtracts from the collector supply voltage, decreasing it.
 - * For that reason, the class C amplifier current pulses are smaller, resulting in a lower-amplitude carrier output.
 - * For 100% modulation the peak of the modulating signal must be equal to V_{ce} , In practice, 100 percent modulation cannot be achieved with the high-level collector modulator circuit. because of the transistor's nonlinear response to small signals.

Principles of Communication Systems

Problems:

- ① An AM transmitter uses high-level modulation of the final RF power amplifier, which has a dc supply voltage V_{cc} of 48V with a total current I of 3.5A. The efficiency is 70 percent.
- (a) what is the RF input power to the final stage.
 - (b) How much AF power is required for 100% modulation.
 - (c) what is the carrier output power?
 - (d) what is the power in one sideband for 67 percent modulation?
 - (e) what is the maximum and minimum dc supply voltage swing with 100 percent modulation?

⇒

Principles of Communication Systems

Solution:

- (a) RF input power or
DC input power or, carrier power P_c

$$P_c = V_{cc} I$$

$$P_c = 48 \times 3.5$$

$$P_c = \underline{168 \text{ W}}$$

(b) $P_m = \frac{P_c}{2} = \frac{168}{2} = 84 \text{ W}$

AF modulating power P_m is one-half the input power.

- (c) carrier output power: $P_{out} = ?$

$$\% \text{ efficiency} = \frac{P_{out}}{P_{in}} \times 100$$

$$P_{out} = \frac{\text{efficiency} \times P_{in}}{100}$$

$$= \frac{70 \times 168}{100} = \underline{1176 \text{ W}}$$



Principles of Communication Systems

- (d) power in one sideband for 67 percent modulation.

$$P_{SB} = \frac{\mu^2 P_c}{4}$$
$$= \frac{(0.67)^2 \times 168}{4}$$

$$P_{SB} = \underline{18.85 \text{ W.}}$$

- (e) maximum and minimum dc supply voltage

minimum swing = 0

maximum swing = $2 \times V_{CC}$

$$= 2 \times 48 = \underline{96 \text{ V}}$$

Principles of Communication Systems

Amplitude Demodulators:

* Demodulators or detectors are circuits that accept modulated signals and recover the original modulating information.

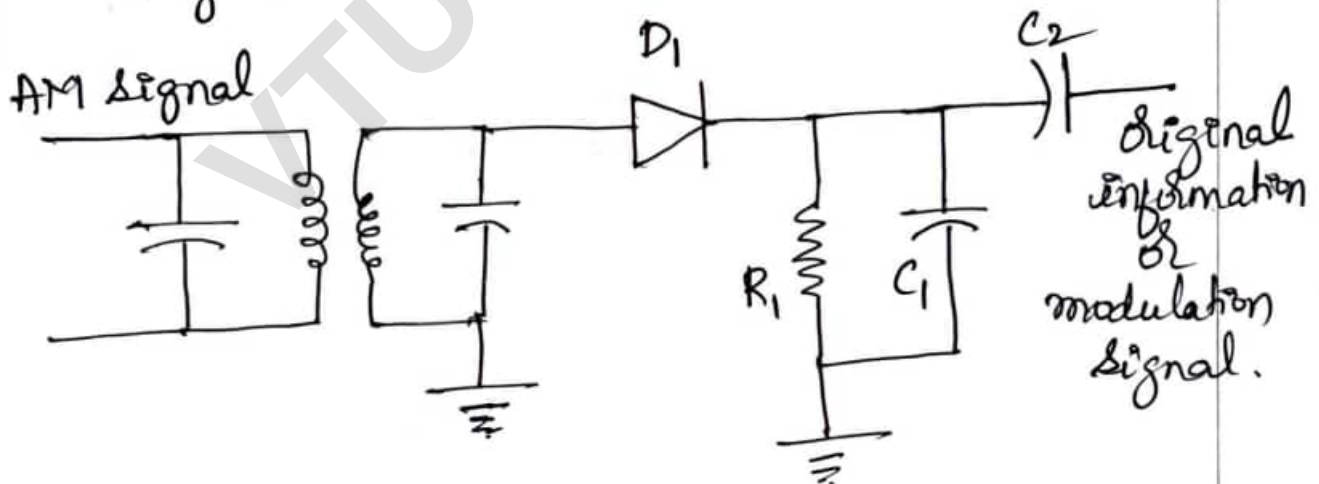
* There are two types of amplitude demodulators:

① Diode Detector.

Diode Detector or Envelope Detector.

* The simplest and most widely used amplitude demodulator is the diode detector.

* A diode detector AM demodulator diagram is as shown in fig.



Principles of Communication Systems

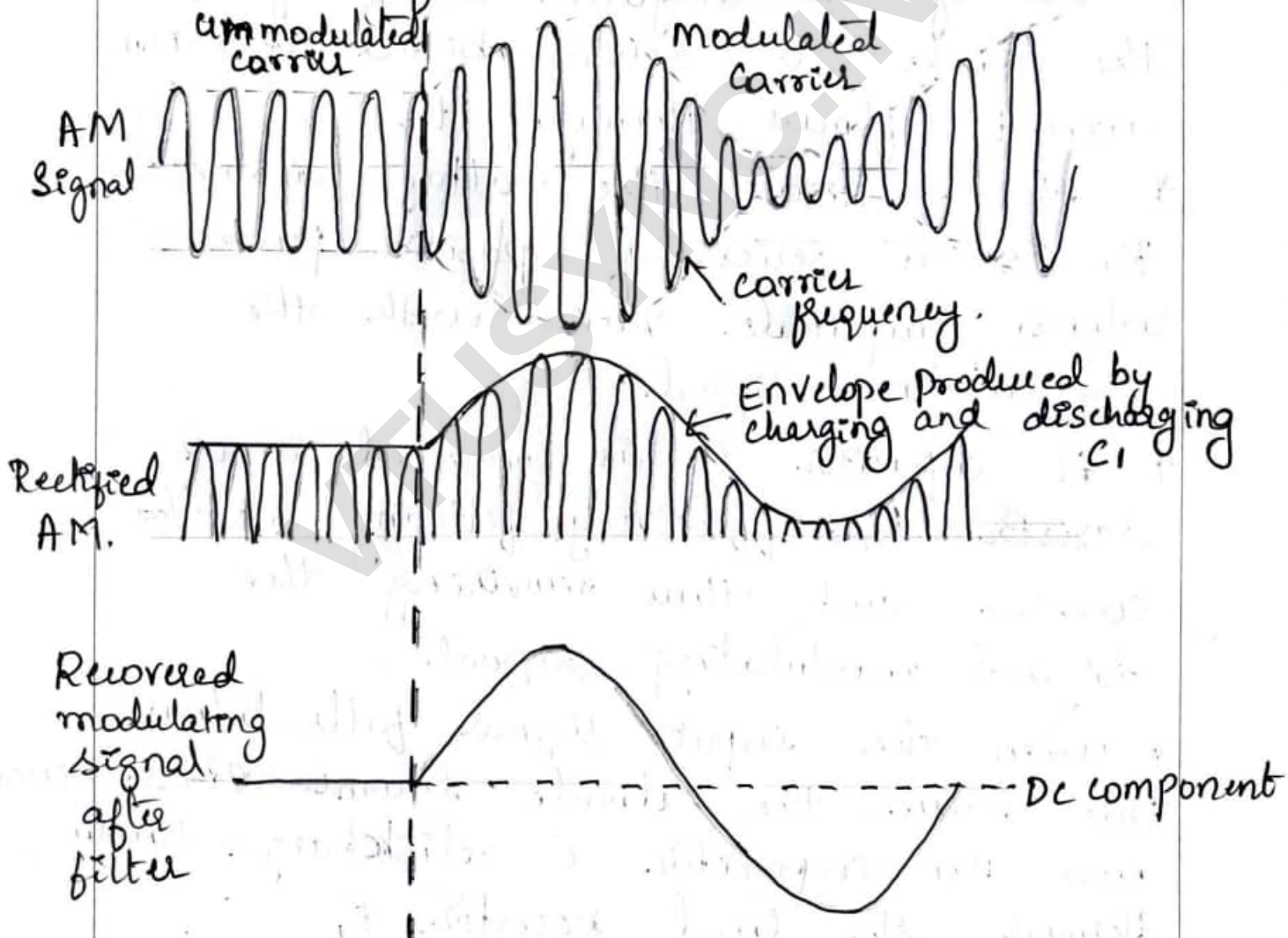
- * As shown in fig, the AM signal is usually transformer-coupled and applied to a basic half-wave rectifier circuit consisting of D_1 and R_1 .
- * The diode conducts when the positive half-cycles of the AM signals occur.
- * During the negative half-cycles, the diode is reverse-biased and no current flows through it.
- * As a result, the voltage across R_1 is a series of positive pulses whose amplitude varies with the modulating signal.
- * A capacitor C_1 is connected across resistor R_1 , effectively filtering out the carrier and thus recovering the original modulating signal.
- * When the input signal falls below this value, the diode becomes reverse-biased and the capacitor C discharges slowly through the load Resistor R_1 .

Principles of Communication Systems

* The discharging process continues untill the next positive half-cycle.

* When the input signal becomes greater than the voltage across the capacitor, the diode conducts again and the process is repeated.

* The waveform is illustrate as below:-



Principles of Communication Systems

Balanced Modulator:

* A balanced modulator is a circuit that generates DSB signal, suppressing the carrier and leaving only the sum and difference frequencies at the output.

* The output of a balanced modulator can be further processed by filters or phase shifting circuitry to eliminate one of the sidebands, resulting in an SSB signal.

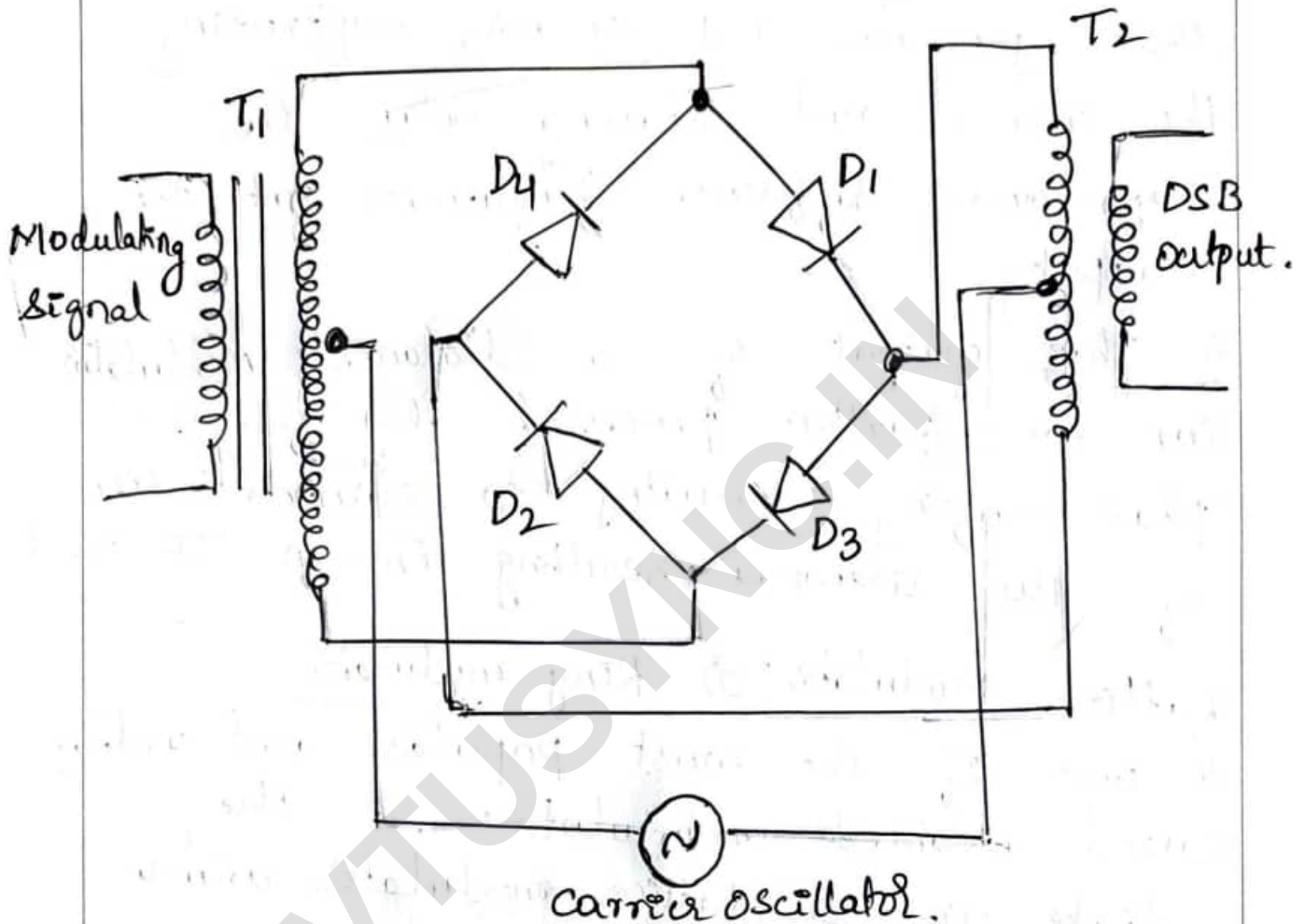
Lattice Modulator: or Ring modulator

* one of the most popular and widely used balanced modulators is the diode ring or lattice modulator which is shown in fig.



Principles of Communication Systems

* Lattice Demodulator.



Carrier oscillator.

* It consists of an input transformer T_1 , an output transformer T_2 , and four diodes connected in a bridge circuit.

* The carrier signal is applied to the center taps of the input and output transformers and the modulating signal is applied to the input transformer T_1 .

Principles of Communication Systems

- * The output appears across the secondary of the output transformer T_2 .
- * The diodes connected in the bridge act like switches, their switching is controlled by the carrier signal.

operation:

① During +ve half cycle of carrier signal.

- * To understand the working of circuit, we first assume that modulating e/p is zero i.e. $m(t) = 0$.
- * During +ve half cycle, D_1 and D_2 are forward biased & D_3 and D_4 are reversed biased.
- * During this, the current divides equally in upper part of the winding produces a magnetic field that is equal and opposite to the magnetic field produced by the current in the lower half of the secondary.

Principles of Communication Systems

* Since magnetic field are equal & opposite, they cancel each other, producing no o/p at the secondary of T_2 .
Thus the carrier is suppressed.

2) During -ve half cycle of the carrier signal:

* During -ve half cycle D_1 and D_2 are reverse biased and diodes D_3 and D_4 are forward biased.

* Similar to +ve half cycle, here also magnetic field are equal & opposite, they cancel each other, producing no o/p at the secondary of T_2 . Thus the carrier is suppressed.

3) with modulating signal: $m(t)$

* Now assume that a low frequency sine wave is applied to the primary of T_1 as the modulating signal.

* During +ve half cycle, the diode D_1 and D_2 are forward biased, the modulating signal at the secondary of T_1 is applied to the primary of T_2 through D_1 & D_2 .

Principles of Communication Systems

* During -ve half cycle D_3 and D_4 are forward biased, the modulating signal at the secondary of T_1 is applied to T_2 with reverse connections.

this results 180° phase shift.

* The Lattice modulator is more ideal when carrier is square wave.

* Wave form illustrating the operation of Lattice modulator for a sinusoidal modulating wave.

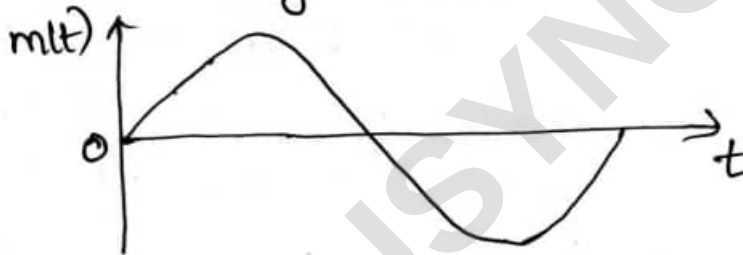


fig (a) modulating signal.

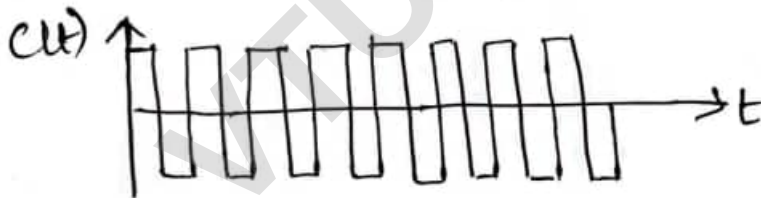


fig (b) carrier signal.

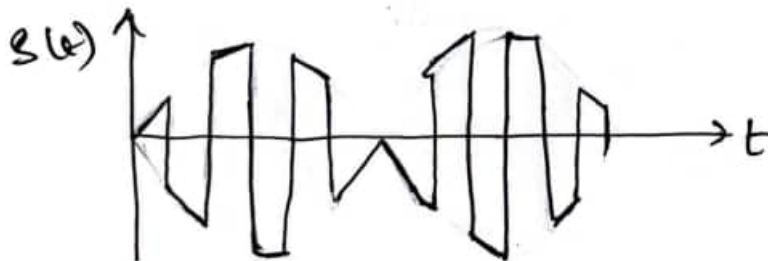


fig (c) DSB-SC modulated wave.

Principles of Communication Systems

Handwritten notes on the page, mostly illegible due to the watermark. The text appears to be a series of paragraphs discussing communication systems, possibly covering topics like modulation, demodulation, and signal processing.

Handwritten notes on the left side of the page, likely describing the first graph.



Handwritten notes on the left side of the page, likely describing the second graph.



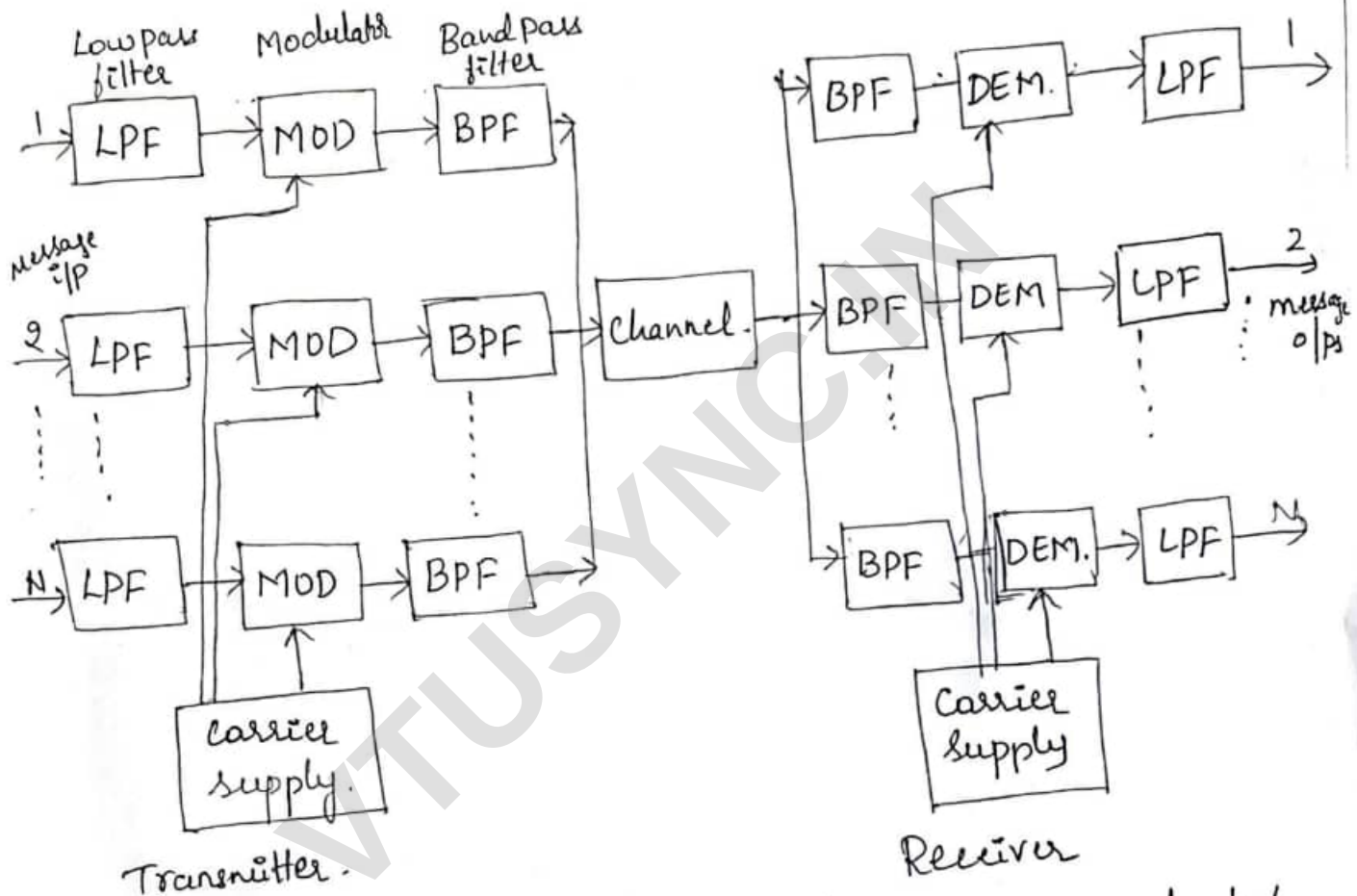
Handwritten notes on the left side of the page, likely describing the third graph.



PRINCIPLES OF COMMUNICATION SYSTEMS -

Frequency Division Multiplexing (FDM).

The block diagram of FDM system is shown in Fig.



- The incoming message signals are assumed to be a low-pass frequency.
- Each signal is passed through an LPF to remove unwanted high frequency components, the filtered signals are applied to modulators that shift the frequency ranges of the signals so to occupy mutually exclusive intervals.

PRINCIPLES OF COMMUNICATION SYSTEMS

- The necessary carrier frequencies is generated from carrier supply, for analog signals SSB modulation is used usually with a bandwidth of 4 K Hz.
- The band pass. filters are used to restrict the band of each modulated wave to its prescribed range.
- The band passed. o/p are combined parallel to form i/p to common channel.
- At Receiver side, a bank of parallel band filters are used to separate the modulated message signals then demodulate uses carrier to get the original message signals.
- To provide two way transmission, we have to completely duplicate the multiplexing facilities with components connected in reverse order and signal waves processing from left to right.